Hyunki Min

Employment

Limited Term Professor, UGA — *2025 - 2026*
Mentor: Gordana Matić

Hedrick Assistant Adjunct Professor, UCLA — *2022 - 2025*
Mentor: Ko Honda

Postdoctoral Associate, MIT — *2021 - 2022*
Mentor: Lisa Piccirillo

Education

Ph.D. Mathematics, Georgia Tech — *2015 - 2021*
Advisor: John Etnyre

B.S. Computer Science and Engineering, Seoul National University — *2008 - 2015*

Visits

Summer Collaborators Program, IAS — *July 2026*
with John Etnyre, Bülent Tosun and Konstantinos Varvarezos

Papers

Tight surgery on torus knots
with John Etnyre, Bülent Tosun and Konstantinos Varvarezos
arXiv:2610.14218

Cable links of uniformly thick knot types
with Rima Chatterjee, John Etnyre and Thomas Rodewald
arXiv:2507.03185

Bordered contact invaraints and half Giroux torsion
with Konstantinos Varvarezos
arXiv:2506.14050

Spinal open books and symplectic fillings with exotic fibers
with Agniva Roy and Luya Wang
arXiv:2410.10697

On contact invariants in bordered Floer homology
with Konstantinos Varvarezos
arXiv:2410.05511

Tight contact structures on a family of hyperbolic L-spaces
with Isacco Nonino
Geometriae Dedicata, 220-45 (2026)

Spaces of Legendrian cables and Seifert fibered links
with Eduardo Fernández
Mathematische Annalen, 394-50 (2026)

Existence and construction of non-loose knots
with Rima Chatterjee, John Etnyre and Anubhav Mukherjee
IMRN, 2025-14 (2025)

Small symplectic caps and embeddings of homology balls in the complex projective plane
with John Etnyre, Lisa Piccirillo and Agniva Roy
arXiv:2305.16207
To appear in *Selecta Mathematica*

The contact mapping class group and rational unknots in lens spaces
IMRN, 2024-15 (2024) 11315–11342

Non-loose torus knots
with John Etnyre and Anubhav Mukherjee
arXiv:2206.14848
To appear in *Memoires of the AMS*

Strongly fillable contact structures without Liouville fillings
arXiv:2205.09912

Cabling Legendrian and transverse knots
with Apratim Chakraborty and John Etnyre
Journal of Differential Geometry, 126-1 (2024) 1-48

On 3-manifolds that are boundaries of exotic 4-manifolds
with John Etnyre and Anubhav Mukherjee
Transactions of the AMS, 375-6 (2022) 4307-4332

Classification of tight contact structures on surgeries on the figure-eight knot
with James Conway
Geometry & Toplogy, 24-3 (2020) 1457–1517

Talks

Topology Seminar, OSU	*April 2026*
Symplectic Seminar, IAS	*April 2026*
AMS Sectional Meeting, Georgia Southern University	*March 2026*
Tech Topology Conference, Georgia Tech	*December 2025*
Topology seminar, UGA	*September 2025*
AMS Sectional Meeting, Cal Poly	*May 2025*
Topology seminar, Caltech	*April 2025*
Topology seminar, Boston College	*March 2025*
Topology seminar, UGA	*November 2024*
Topology seminar, KIAS	*October 2024*
Topology seminar, University of Arkansas	*May 2024*
Princeton Topology Conference, Princeton University	*April 2024*
Australian Geometric Tropology Webinar, Zoom	*Mar 2024*
Topology seminar, University of Southern California	*November 2023*

	Topology seminar, Claremont Colleges	*November 2023*
	Topology seminar, UT Austin	*September 2023*
	Graduate Student Topology and Geometry Conference, Harvard University	*July 2023*
	Georgia Topology Conference, University of Georgia	*May 2023*
	AMS Sectional Meeting, Georgia Tech	*March 2023*
	Workshop for Young Symplectic Geometers, Korea	*February 2023*
	G&T Workshop Turkey, Online workshop	*December 2021*
	Boston Graduate Topology Seminar, MIT	*November 2021*
	Informal Symplectic Seminar, MIT	*November 2021*
	Geometry/Topology Seminar, MIT	*September 2021*
	GT GAPS, Online seminar	*August 2021*
	Joint Mathematics Meeting, Online conference	*January 2021*
	Tech Topology Conference, Georgia Tech	*December 2020*
	NCNGT 20, Online conference	*June 2020*
	Topology Seminar, University of Alabama	*February 2020*
	Topology Seminar, Seoul National University	*December 2019*
	Tech Topology Conference, Georgia Tech	*December 2019*
	Topology Seminar, Binghamton University	*April 2019*
	AMS Sectional Meeting, University of Auburn	*March 2019*
	Topology Seminar, University of Georgia	*January 2019*
	Mini-workshop, Seoul National University	*January 2019*
	Tech Topology Conference, Georgia Tech	*December 2018*

Teaching

UGA	**Math 2700: Differential Equations**	*Spring 2026*
	Math 2260: Calculus II	*Fall 2025*
UCLA	**Math 32B: Multivariable Calculus**	*Spring 2025*
	Math 120A: Differential Geometry	*Winter 2025*
	Math 32B: Multivariable Calculus	*Winter 2025*
	Math 132: Complex Analysis for Applications	*Fall 2024*

	Math 132: Complex Analysis for Applications	*Spring 2024*
	Math 110B: Algebra	*Winter 2024*
	Math 132: Complex Analysis for Applications	*Winter 2024*
	Math 31B: Integration and Infinite Series	*Fall 2023*
	Math 33A: Linear Algebra and its Applications	*Spring 2023*
	Math 33B: Differential Equations	*Winter 2023*
	Math 115A: Linear Algebra	*Fall 2022*
MIT	**TA of 18.03: Differential Equations**	*Spring 2022*
	TA of 18.02: Multivariable Calculus	*Fall 2021*
Georgia Tech	**GRE Math Boot Camp**	*Summer 2021*
	Math 2550: Introduction to Multivariable Calculus	*Summer 2020*
	Math 2550: Introduction to Multivariable Calculus	*Summer 2019*
	Math 2551: Multivariable Calculus	*Summer 2018*
	Head TA of Math 2551: Multivariable Calculus	*Spring 2021*
	Head TA of Math 2551: Multivariable Calculus	*Fall 2019*
	TA of Math 1554: Linear Algebra	*Fall 2020*
	TA of Math 2551: Multivariable Calculus	*Spring 2020*
	TA of Math 2551: Multivariable Calculus	*Spring 2019*
	TA of Math 1554: Linear Algebra	*Fall 2018*
	TA of Math 2552: Differential Equations	*Fall 2017*
	TA of Math 2551: Multivariable Calculus	*Summer 2017*
	TA of Math 2551: Multivariable Calculus	*Spring 2017*
	TA of Math 1564: Linear Algebra with Abstract Vector Spaces	*Fall 2016*
	TA of Math 2552: Differential Equations	*Summer 2016*
	TA of Math 2551: Multivariable Calculus	*Spring 2016*

Awards

O'Neill Travel Award, UCLA	*Spring 2025*
Top Grad Student Award, Georgia Tech	*Spring 2021*
Outstanding Grad Student Instructor Award, Georgia Tech	*Spring 2021*

Outstanding TA Award, Georgia Tech *Spring 2021*

Mentoring

REU: 2-bridge knots and the cosmetic surgery conjecture, UCLA *Summer 2023*
Mentees: Michael Huang, Zelong Li, Rahi Tanaz, Chengyi Zhang

Direct Reading Program: Legendrian knots, MIT *Winter 2021*
Mentees: Pedro Suarez, Liane Xu

Service

Referee work: *Algebraic and Geometric Topology, Asian Journal of Mathematics, Bullentins of London Mathematical Society, Journal of Differential Geometry, Geometry & Topology, Journal of European Mathematical Society, Journal of Knot Theory and its Ramifications, Journal of Symplectic Geometry, Journal of Topology, New York Journal of Mathematics, Topology and its Applications, Transactions of the AMS*

Co-organizer, Geometry & Topology Seminar, UCLA *Spring 2024*

Co-organizer, Special Session at AMS Sectional Meeting, UC Riverside *Fall 2024*

Co-organizer, Geometry & Topology Seminar, UCLA *Fall 2024*

Co-organizer, Geometry & Topology Seminar, MIT *Spring 2022*

Co-organizer, Geometry & Topology Seminar, MIT *Fall 2021*

Co-organizer, Contact and symplectic topology, NCNGT 21 *Summer 2021*

Co-organizer, Reading groups on filling obstructions, Georgia Tech *Spring 2021*

Co-organizer, 4-dimensional manifolds reading seminar, Georgia Tech *Fall 2020*

Co-organizer, Heegaard Floer homology reading seminar, Georgia Tech *Spring 2020*

Organizer, High dimensional contact topology reading seminar, Georgia Tech *Fall 2019*

Grad Panel with Visiting Honor Students, Georgia Tech *Spring 2021*

References

John Etnyre, Georgia Tech

Jen Hom, Georgia Tech

Ko Honda, UCLA

Lisa Piccirillo, UT Austin

Klara Grodzinsky (Teaching), Georgia Tech

Guido Montufar (Teaching), UCLA